Predict New York City Airbnb Prices

Project Summary — Linear Regression

Objective

The objective of this project is to build and evaluate a Linear Regression model that predicts Airbnb listing prices in New York City using available listing features.

Data Preparation

The dataset was inspected for its shape, columns, data types, and missing values. The *last_review* column and identifier/text columns (*id*, *name*, *host_id*, and *host_name*) were removed for this model. Missing *reviews_per_month* values were filled with 0 after checking that those rows had zero reviews. Listings above the 95th-percentile limits for price ($355) or minimum nights (30) were filtered out. The high-cardinality *neighbourhood* column was removed, and *neighbourhood_group* and *room_type* were one-hot encoded using *drop_first=True*.

Model Development

After preprocessing, the dataset contained 45,759 rows. The target variable was *price*, and the remaining 13 columns were used as features. Data was split into 80% training data (36,607 rows) and 20% testing data (9,152 rows), using *random_state=42*. A Linear Regression model was trained on the training set and evaluated on the unseen test set.

Evaluation Results

Metric	Result	Interpretation
R^2 score	0.471	Explains about 47.1% of price variation in the test set.
MAE	38.54	Average absolute prediction error is about 38.54 price units.
MSE	2,834.96	Mean squared prediction error.
RMSE	53.24	Root mean squared error, in price units.

Visual and Group Analysis

The actual-versus-predicted scatter plot showed a general upward relationship but substantial scatter, indicating imperfect predictions. The residual plot showed errors on both sides of zero and a changing spread, suggesting the model does not capture all price patterns. When listings were divided at the test-set median actual price of $100, the MAE was 29.03 for listings priced at or below $100 and 48.58 for listings above $100. Thus, the model predicted cheaper listings more accurately by MAE.

Conclusion

The Linear Regression model provides a useful baseline but has moderate predictive performance ($R^2 \approx 0.47$). It performs better for cheaper listings than for expensive ones. Future work could explore additional informative features, alternative regression models, and further validation to improve accuracy.